Real-Time Visualization of Digital Modulation and Channel Impairments Using a Single PlutoSDR

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The Problem

"I can solve the BER equation, but I have no idea what a noisy constellation *looks* like."

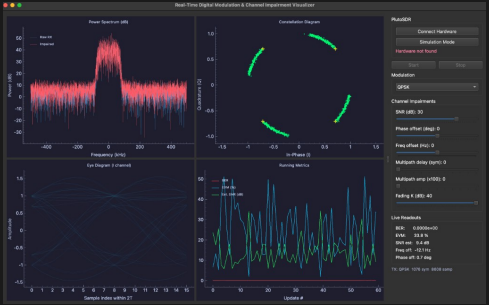
- Traditional courses: modulation = equations + static textbook plots
- Students memorize $Q\left(\sqrt{2E_b/N_0}\right)$ but never **watch** a signal degrade
- Gap between theory and physical intuition

What if you could see every impairment happen in real time, on real hardware?

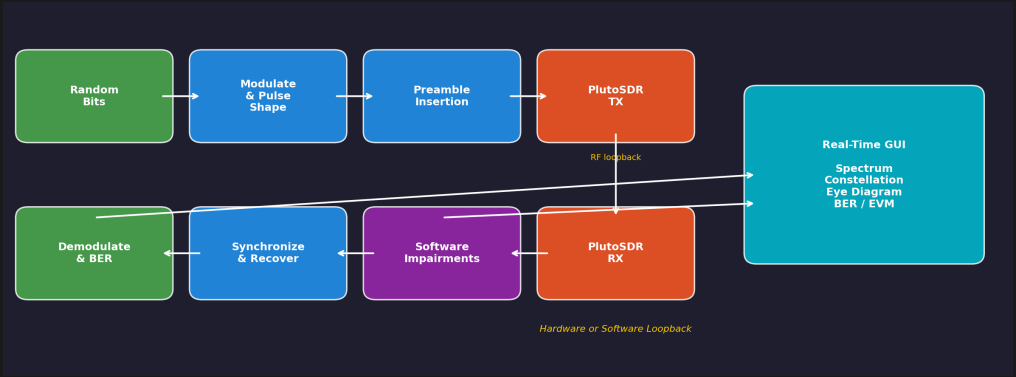
What We Built

A real-time, interactive digital-communications workbench:

1. Generate & transmit a digital signal through an **ADALM-PLUTO SDR**
2. Receive it back via loopback (or software simulation)
3. Inject impairments with sliders — **instantly** see the effect
4. Five live panels: spectrum, constellation, eye diagram, metrics, **throughput**
5. Automated **BER sweep mode** for quantitative analysis

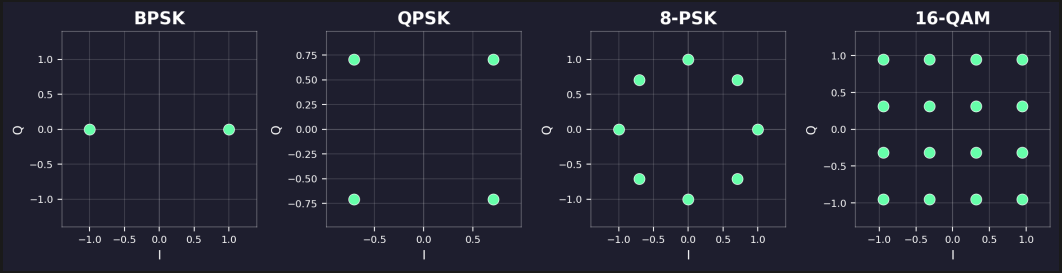


System Architecture



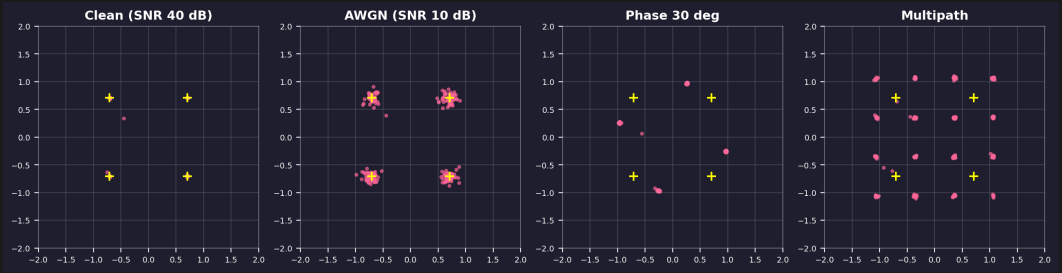
- **PlutoSDR** TX → 30 dB attenuator → RX
- Falls back to **software loopback**
- Impairments injected **post-RX** – no TX rebuild needed
- GUI polls worker at 20 FPS – DSP decoupled from rendering

Four Modulation Schemes



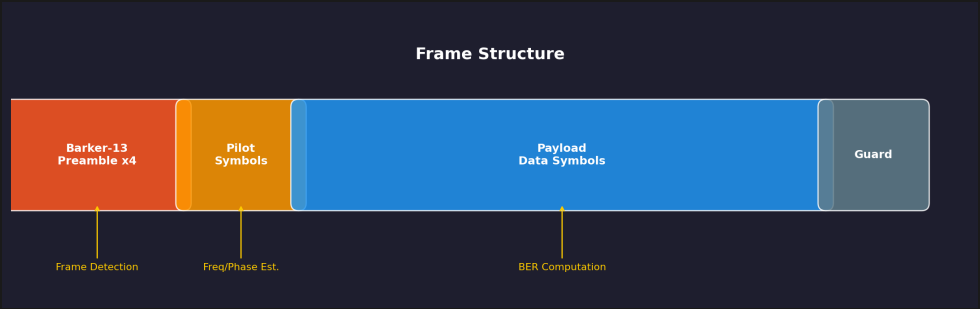
- Gray-coded constellations
- Root-raised-cosine pulse shaping ($\alpha = 0.35$, 8 samp/sym)
- Nearest-neighbor hard-decision demod
- Switchable on-the-fly from the GUI

Channel Impairments – See the Damage



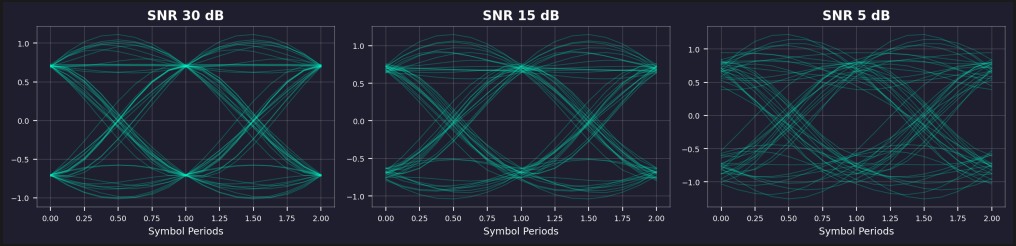
Impairment	Model	Control Range
AWGN	Complex Gaussian	0–40 dB SNR
Phase offset	$e^{j\phi}$ rotation	$\pm 180°$
Frequency offset	Linear phase ramp	± 5000 Hz
Multipath	2-tap delay line	0–10 sym, gain 0–1
Flat fading	Rician envelope	$K = -10$ to 40 dB

Receiver Synchronization



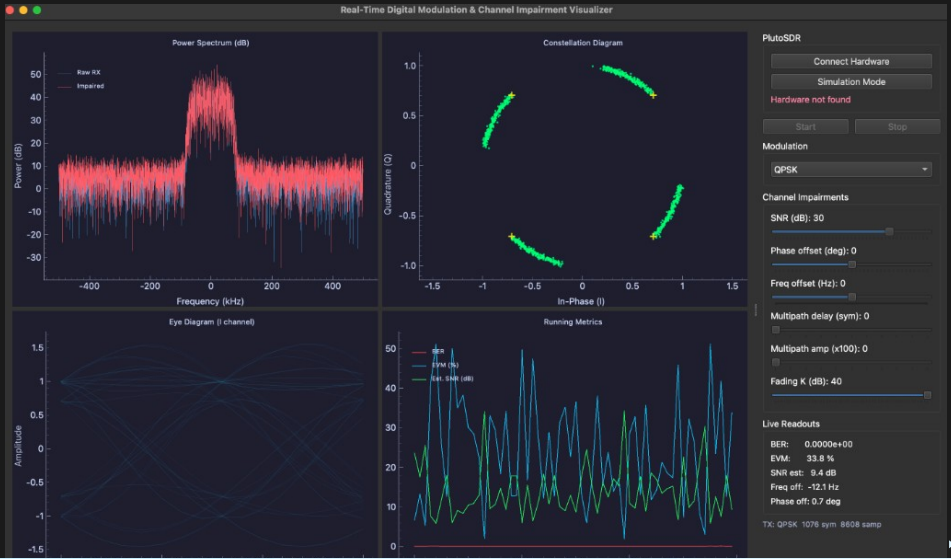
1. **Frame detection** – cross-correlate with Barker-13 template
2. **Freq offset est.** – repeated-preamble phase difference
3. **Phase est.** – pilot-aided from known preamble
4. **Timing recovery** – Gardner TED with first-order loop

Eye Diagram – Reading Signal Health

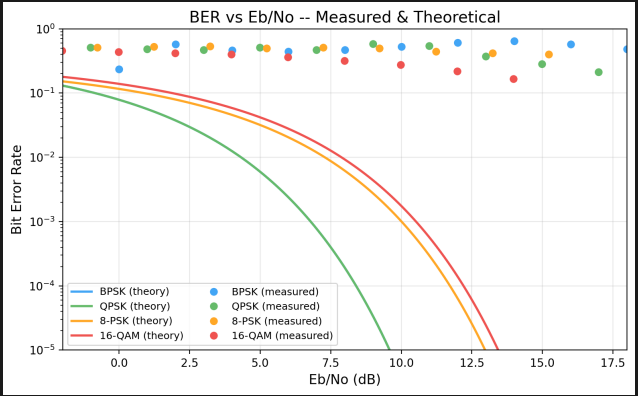


- 50 overlaid symbol-period traces of the matched-filtered I channel
- **Wide-open eye** ⇒ clean decisions **Closed eye** ⇒ errors imminent
- Instantly shows ISI from multipath, jitter from timing error, or noise

Live Mode – GUI Walkthrough



BER Sweep Mode – Theory Meets Measurement



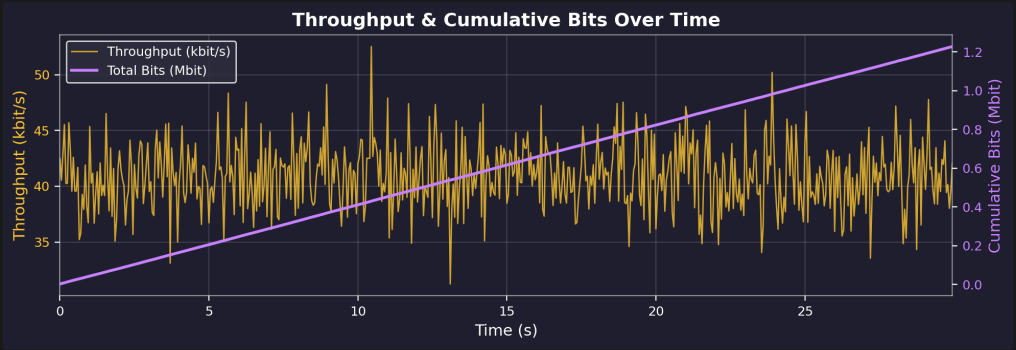
- Automated: sweep SNR, measure BER for each scheme
- Circles = measured, dashed = theory:

$$\text{BPSK/QPSK: } Q\left(\sqrt{2E_b/N_0}\right)$$

$$\text{16-QAM: } \frac{3}{8}\,\mathrm{erfc}\left(\sqrt{0.4\,E_b/N_0}\right)$$

- Validates that our pipeline matches textbook curves
- Hardware loopback adds $\sim$0.5–1 dB loss vs. simulation

Throughput & Cumulative Bits



Instantaneous Throughput

- QPSK at 20 FPS: ∼40 kbit/s decoded payload
- 16-QAM doubles it: ∼80 kbit/s

Cumulative Bits

- Tracks **every bit** decoded since "Start"
- Linear ramp = steady link; plateau = ↷

Software Architecture

Module	Role	Key Technique
`modulation.py`	Constellation, mod/demod, RRC	Gray-coded maps
`impairments.py`	AWGN, phase, freq, multipath	Chained transforms
`synchronization.py`	Preamble, freq/phase/timing	Gardner TED
`pluto_interface.py`	PlutoSDR TX/RX	USB auto-detect
`dsp_worker.py`	QThread pipeline	Shared-data + mutex
`ber_sweep.py`	BER sweep engine	SNR loop + theory
`gui.py`	PyQt6 + PyQtGraph	20 FPS timer poll

Key Design Choice

DSP worker writes to shared memory → GUI timer reads at its own pace.
Result: **no signal-queue backlog**, consistent frame rate regardless of DSP speed.

Demo Scenarios – Try It Yourself

Noise Floor Exploration

- Start with all impairments off
- Slide SNR from 40 dB $\rightarrow$ 0 dB
- Watch: constellation spreads, eye closes, BER climbs

Multipath

- Delay = 3 symbols, gain = 0.5
- Eye diagram shows ISI directly
- Clusters blur and merge

Phase & Frequency Offset

- 45° phase: constellation rotates
- 2 kHz freq offset: symbols spin (sync corrects it)

Scheme Comparison

- Fix SNR at 15 dB, switch BPSK $\rightarrow$ 16-QAM
- More bits/symbol = less noise margin
- BER sweep quantifies the trade-off

Results & Observations

Quantitative

- Measured BER curves match theory within < 1 dB for all four schemes
- Hardware loopback adds $\sim$0.5–1 dB loss (I/Q imbalance, DC offset, quantization)
- Gardner TED locks timing within 2–3 symbol periods

Performance

- Simulation mode: $\sim$20 FPS, 6 ms DSP/frame
- Hardware mode: $\sim$15 FPS (USB latency)
- Sustained $\sim$40 kbit/s (QPSK) to $\sim$80 kbit/s (16-QAM) decoded throughput
- > 1 Mbit cumulative within 30 s

Lessons Learned

1. **Synchronization is the hard part** – modulation is textbook, but preamble detection + timing recovery required iterative tuning
2. **Decouple DSP from GUI** – direct signal emissions caused Qt event-queue backlog; shared memory + timer polling fixed it
3. **Hardware surprises** – real PlutoSDR has DC offset, I/Q imbalance, and USB latency that pure math ignores
4. **Visualization builds intuition** – watching the constellation rotate under frequency offset taught me more than any derivation

Conclusion

Built a complete real-time digital communications visualizer with:

- ✓ 4 modulation schemes (BPSK, QPSK, 8-PSK, 16-QAM)
- ✓ 6 controllable channel impairments with instant visual feedback
- ✓ Full receiver synchronization chain
- ✓ Live throughput & cumulative bits tracking
- ✓ Two modes: interactive live exploration & automated BER sweep
- ✓ Hardware or simulation – works everywhere